Beating SPY with an Integrated AI Trading Strategy: FinRL-Trading, FinRobot, and FinGPT

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1. Introduction

The goal of this project is to build an integrated AI-driven trading strategy that combines quantitative portfolio construction, analyst-style equity research, and financial news sentiment signals. Instead of relying on a single model, our strategy connects three components from the AI4Finance ecosystem:

- **FinRL-Trading** for stock selection, ranking, portfolio construction, and backtesting.
- **FinRobot** for equity research, thesis validation, valuation analysis, catalyst analysis, and risk assessment.
- **FinGPT** for financial news sentiment and short-horizon exposure-control signals.

The assignment requires the final strategy to be evaluated against **SPY** over the exact backtest window from **2026–01–01 to 2026–04–30**, and to explain how stocks are selected, how signals are combined, how weights are determined, and what each module contributes.

Our final design follows a **quant-first, LLM-confirmation** structure. FinRL-Trading first builds a base portfolio. FinRobot then evaluates the selected names using point-in-time research snapshots and adjusts the base weights through structured research flags. FinGPT is tested as a separate news-sentiment module to determine whether short-horizon financial news can provide useful exposure-control information.

The key result is that our integrated strategy outperformed SPY during the required window. In our Attempt 7 integrated backtest, the FinRL-only portfolio returned **17.09%**, the FinRL + FinRobot overlay returned **17.87%**, and SPY returned **5.19%**. The FinRobot overlay improved the FinRL baseline by **0.78 percentage points** and outperformed SPY by **12.68 percentage points**.

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2. System Architecture

Our system is built around a layered pipeline:

```
FinRL-Trading
Adaptive Rotation Strategy
Base stocks + base weights
        ↓
FinRobot
Point-in-time equity research snapshots
Risk / valuation / catalyst / sentiment / thesis
        ↓
Rule-Based Portfolio Overlay
Flag-to-multiplier conversion
Adjusted portfolio weights
        ↓
FinGPT Sentiment Module
Weekly Alpaca news sentiment test
Long-only exposure filter
        ↓
Final Evaluation
Backtest vs SPY
```

The main principle is that each module has a distinct role:

ComponentRoleFinRL-TradingSelects candidate assets, ranks them, constructs weekly base portfolio weights, and runs the main backtest.FinRobotPerforms equity research and investment thesis validation on the selected names, then converts research into structured weight adjustment signals.FinGPTTests whether weekly financial news sentiment can provide additional short-horizon exposure-control information.

This design avoids using LLMs as black-box portfolio managers. Instead, we use LLM-based modules to produce structured and auditable signals that can be translated into deterministic portfolio rules.

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3. FinRL-Trading: Adaptive Rotation Strategy

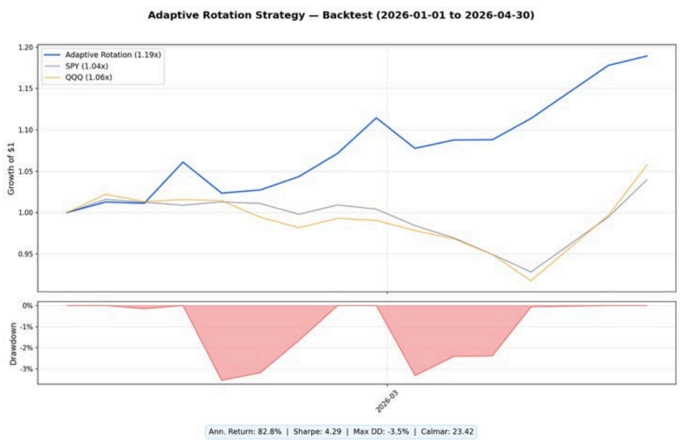
The FinTrading component is the quantitative foundation of the system. It uses an Adaptive Rotation Strategy to generate the initial portfolio. The strategy incorporates market regime detection, asset group ranking, intra-group ranking, risk controls, and portfolio construction.

When reproducing the Adaptive Rotation Strategy, we added a **QQQ rolling selection enhancement** to the stock-selection layer. Specifically, we introduced a momentum-based filtering step that ranks assets using a combination of **6-month and 12-month momentum**, while excluding the most recent period to reduce short-term reversal noise. This rolling momentum score acts as an initial screening mechanism, narrowing the candidate pool before applying the original residual momentum ranking.

We further improved robustness by incorporating a **volatility-adjusted scoring mechanism**. Instead of selecting stocks only by raw momentum, we penalized high-volatility assets so that the strategy favors more stable momentum signals. This helps reduce the risk of chasing unstable short-term winners.

During the backtest period from **2026–01–01 to 2026–04–30**, the Adaptive Rotation Strategy remained highly invested on average, with an average invested weight of about **93.56%**, and held approximately **5.6 assets**. The regime distribution showed that the market was mostly **risk-on**, with risk-on conditions appearing in about **73.3%** of the period, fast risk-off in **20.0%**, and risk-off in **6.7%**.

Figure 1. Adaptive Rotation Strategy Backtest, 2026–01–01 to 2026–04–30.



The Adaptive Rotation Strategy achieved approximately **1.19x cumulative growth**, outperforming both SPY and QQQ in the same period. The chart also shows a relatively small drawdown profile, with key metrics including an annualized return of about **82.8%**, Sharpe ratio of **4.29**, maximum drawdown of about **-3.5%**, and Calmar ratio of **23.42**.

Overall, the FinRL-Trading module provided the core alpha engine. It selected the portfolio, captured the main market opportunity, and supplied the base weights used by the rest of the system.

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4. FinRobot: Research-Based Validation and Portfolio Overlay

The FinRobot component was added as a research-based validation layer on top of the FinTrading portfolio. Its purpose was not to independently select stocks or replace the quantitative strategy. Instead, FinRobot consumed the stocks and weights produced by FinRL-Trading, generated point-in-time research snapshots, extracted structured research flags, and used those flags to adjust the original portfolio weights.

The integrated FinRobot workflow was organized around `integrated_backtest.py`. The workflow normalizes FinRL base weights, generates or reuses FinRobot research snapshots, extracts structured research flags with an LLM, applies deterministic overlay rules, and writes adjusted weights, audit files, equity curves, and metrics.

4.1 Research Universe

FinRobot does not create a new stock universe. Stock selection remains the responsibility of the FinRL base workflow. The overlay reads the FinRL output weights and applies research-driven adjustments to names that appear in the base portfolio or research universe.

The main researched tickers in Attempt 7 were:

```
AAPL, BHP, COP, CVX, FCX, GOOGL, NVDA, TSLA, XOM
```

The broader portfolio also included ETF or defensive assets such as:

```
SPY, QQQ, XLU, XLV, IAU, GLD, IEF, SHY, SLV, CAS
```

This setup prevents FinRobot from becoming a manual stock picker. Instead, it acts as a validation and adjustment layer for the portfolio that the trading model already selected.

4.2 Point-in-Time Research Snapshots

To reduce look-ahead bias, the FinRobot research snapshots are generated in a point-in-time manner. In Attempt 7, research was generated monthly, while FinRL base weights updated weekly. A research snapshot from the first research date of a month was reused as the overlay signal for later weekly rebalances until the next monthly research update.

This creates a practical tradeoff. Monthly research is cheaper and easier to automate, but it can lag fast-moving market conditions. A future version could test weekly FinRobot research snapshots, but that would increase LLM and data costs.

4.3 Structured Research Flags

The key design choice was to transform qualitative research into actionable portfolio signals. For each ticker and research date, the LLM extracted five structured flags:

```
risk_flag: low / medium / high
valuation_flag: cheap / fair / expensive / extre
catalyst_flag: negative / neutral / positive
sentiment_flag: negative / mixed / positive
thesis_strength: weak / moderate / strong
```

These flags allow us to translate analyst-style research into measurable signals. For example, a stock with strong thesis strength, positive catalysts, and manageable risk may receive a higher multiplier, while a stock with high risk, extreme valuation, or weak thesis may be reduced.

Importantly, the LLM does not directly assign portfolio weights. The LLM only extracts flags; deterministic rules then translate those flags into weight multipliers. This keeps the portfolio construction process explainable and reproducible.

4.4 Weight Overlay Rules

The overlay starts from FinRL base weights. It then calculates a multiplier from the extracted FinRobot flags and applies guardrails to avoid excessive LLM-driven changes. Attempt 7 used controls such as a no-op band, maximum relative adjustment, defensive thresholds, and limits on how much cash or ETF allocation could fund stock boosts.

The rule interpretation is straightforward:

```
Positive research flags  → modestly increase or
Neutral research flags   → keep weight mostly un
Negative research flags  → reduce weight and mov
```

This overlay is intentionally conservative. The goal is not to let the LLM dominate the trading model, but to let research evidence refine the base portfolio.

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5. FinGPT: News-Sentiment Exposure Filter

In addition to the monthly FinRobot research overlay, we explored a separate FinGPT-style news-sentiment signal module. The purpose was to test whether short-horizon financial news could provide useful exposure-control information.

While FinRobot focuses on fundamental and analyst-style research snapshots, the FinGPT module focuses on weekly Alpaca news and attempts to extract more timely market sentiment signals. The goal was not to replace the FinTrading allocation engine, but to evaluate whether LLM-derived news sentiment could serve as an additional risk filter or portfolio overlay.

For this module, we constructed a weekly sentiment dataset using Alpaca news for seven selected assets:

```
GLD, SLV, IAU, AMZN, AAPL, COP, XLU
```

The sentiment pipeline uses Agent 1 to classify news sentiment and event type, and Agent 2 to generate signal probabilities or final trading directions. The backtest result files include sentiment labels, sentiment scores, event types, signal probabilities, final direction, realized return, position, and strategy return.

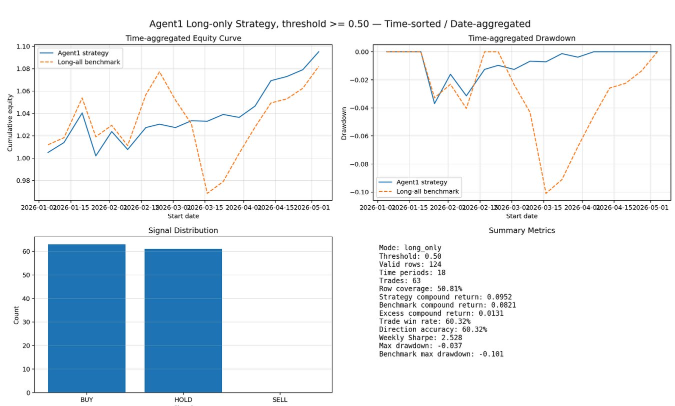
The final sentiment strategy used a simple long-only exposure filter:

```
If positive sentiment probability ≥ 0.50:
    Enter long position
Else:
    Hold
```

Under the time-aggregated weekly backtest, the FinGPT long-only strategy achieved a compound return of approximately **9.52%**, compared with **8.21%** for the long-all benchmark. More importantly, it reduced maximum drawdown from approximately **-10.09%** for the benchmark to approximately **-3.69%**, while maintaining a trade win rate of approximately **60.32%**.

We also tested a mixed long-short version. The mixed version was less attractive, achieving a lower return of around **7.70%** and a lower win rate. This suggests that negative sentiment is better interpreted as a **risk-off signal** rather than a direct short signal.

Figure 2. FinGPT Long-Only and Mixed Long-Short Sentiment Backtests.



The long-only version performed better than the mixed long-short version. This indicates that the model's positive sentiment signal is more useful for confirming long exposure, while negative sentiment should be used mainly to reduce or avoid exposure.

Overall, the sentiment module did not support the original goal of building a standalone LLM-driven trading strategy. However, it showed value as a conservative long-exposure filter. The fine-tuned 8B model was not reliable enough to generate full trading decisions, but its positive sentiment probability helped identify periods where long exposure was more favorable and helped reduce downside capture during volatile market conditions.

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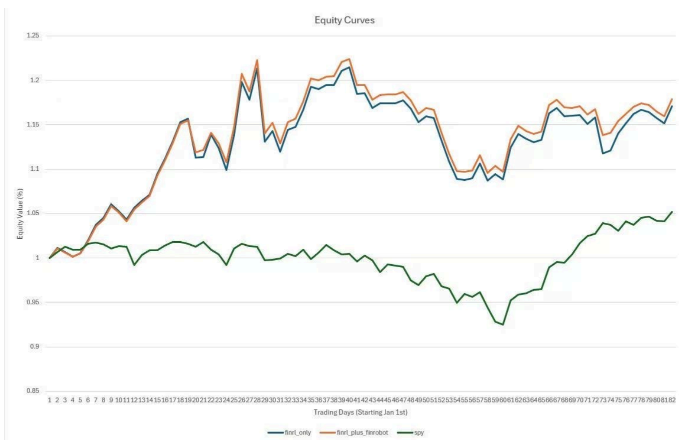
6. Backtest Results

The final integrated strategy was evaluated from **2026–01–01 to 2026–04–30**, matching the required assignment window. The required benchmark was **SP**



The blue line represents the original FinRL-only portfolio, the orange line represents the FinRL portfolio adjusted by the FinRobot research overlay, and the green line represents SPY. The FinRobot-adjusted strategy closely follows the FinRL baseline but remains slightly above it for most of the backtest period, suggesting that the research overlay modestly improved the base quantitative strategy while preserving its overall return profile.

The main performance table is:

StrategyTotal ReturnAnnualized ReturnAnnualized VolatilityMax DrawdownSharpeWin RateFinRL only17.09%62.95%27.44%-10.48%1.9361.73%FinRL + FinRobot overlay17.87%66.34%26.36%-10.48%2.0760.49%SPY5.19%16.97%14.32%-9.13%1.1751.85%

The integrated FinRL + FinRobot strategy outperformed both SPY and the FinRL-only baseline. The return improvement over FinRL-only was modest, but positive:

```
FinRL + FinRobot return delta vs FinRL only: +0.
FinRL + FinRobot return delta vs SPY: +12.68 per
```

The overlay also improved annualized return from **62.95%** to **66.34%**, improved Sharpe ratio from **1.93** to **2.07**, and reduced annualized volatility from **27.44%** to **26.36%**. Max drawdown remained the same as the FinRL-only baseline.

This result shows that most of the performance came from the FinRL-Trading base strategy. However, FinRobot added a small but meaningful improvement in risk-adjusted performance and made the portfolio more explainable.

7.1 FinRL-Trading Contribution

FinRL-Trading was the main source of performance. It selected the investable names, generated weekly base portfolio weights, and captured the main market opportunity during the test period. The Adaptive Rotation Strategy's rolling momentum filter and volatility-aware ranking helped the strategy outperform SPY and QQQ in the standalone FinTrading backtest.

The base FinRL-only strategy was already strong, so the overlay's job was not to create returns from scratch. Instead, the research and sentiment layers needed to improve or preserve the base strategy without damaging its core signal.

7.2 FinRobot Contribution

FinRobot supplied company-level research snapshots, financial analysis, news summaries, and report artifacts. These reports were converted into structured flags and used to adjust portfolio weights.

In Attempt 7, the FinRobot flag distribution was:

Flag TypeDistributionrisk_flagmedium: 23, high: 13valuation_flagfair: 20, expensive: 10, extreme: 4, cheap: 2catalyst_flagneutral: 23, positive: 11, negative: 2sentiment_flagmixed: 29, negative: 4, positive: 3thesis_strengthmoderate: 25, weak: 8, strong: 3

Ticker-level observations helped explain portfolio decisions. GOOGL was the strongest consistently positive name, receiving strong thesis, positive sentiment, and positive catalyst flags. TSLA was consistently weak, with repeated high-risk, weak-thesis, negative-sentiment, and extreme-valuation flags. FCX was also consistently high risk, while AAPL was generally moderate but expensive, with positive catalysts in two months.

This shows that FinRobot contributed both performance refinement and explainability. Each portfolio adjustment can be traced back to research evidence rather than being treated as a black-box trading decision.

7.3 FinGPT Contribution

The FinGPT sentiment module was useful as a separate validation experiment. It showed that positive news sentiment can help identify favorable long-exposure periods and reduce drawdown. However, it was not strong enough to serve as the core alpha engine.

The main insight is that the FinGPT signal is better used as a **risk filter** than as a standalone trading system:

```
Positive sentiment → allow or confirm long expos
Neutral sentiment  → hold
Negative sentiment → reduce risk, not necessaril
```

This is consistent with our mixed long-short test, where short signals produced weaker results than the long-only strategy.

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8. Failure Analysis

8.1 News Quality Is Still Noisy

The news and catalyst layer became operational in Attempt 7, but it still contained noise. Some news coverage was heavily concentrated in AAPL, and some articles were false positives, such as Apple iSports articles that were unrelated to Apple Inc. Some institutional holding articles were also over-classified as acquisition-type catalysts.

Future improvements should include stricter company-name matching, filtering of unrelated symbols, downweighting routine 13F-style ownership updates, and improving the catalyst taxonomy.

8.2 The FinRobot Overlay Is Conservative

The FinRobot overlay improved the FinRL baseline by **0.78 percentage points**, but many raw LLM decisions were neutralized by guardrails. The raw LLM decisions included more cuts and boosts, while the final decisions were mostly keep decisions after applying no-op bands, relative adjustment caps, and defensive guards.

This conservative design reduced the risk of overfitting, but it may also have limited the upside from high-conviction research signals. Future work could test a looser no-op band, higher boost funding limits, or more flexible redeployment of cash released from weak names.

8.3 Monthly Research Can Lag Weekly Trading

FinRL weights update weekly, but FinRobot research snapshots update monthly. This means that the overlay may use stale research during later weeks of a month. A future version should test weekly research frequency, although that would increase API usage and LLM cost.

8.4 FinGPT Is Not Yet Fully Integrated into the Final Portfolio

The FinGPT module was tested as a separate sentiment backtest. It demonstrated potential as a long-only exposure filter, but it was not yet fully merged into the final FinRL + FinRobot portfolio. A more robust future version would combine small-model news filtering, stronger-model sentiment understanding, and a final market-aware ML filter that incorporates price momentum, volatility, regime information, and transaction costs.

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9. Conclusion

Our final system connects FinRL-Trading, FinRobot, and FinGPT into a coherent AI trading workflow.

FinRL-Trading provided the core quantitative engine. It selected the portfolio, generated base weights, and captured most of the strategy's performance. FinRobot added a research-based validation layer, converting company-level analysis into structured flags and deterministic weight adjustments. FinGPT tested whether short-horizon financial news sentiment could provide useful exposure-control signals.

The final integrated result was positive. During the required 2026–01–01 to 2026–04–30 backtest window, the FinRL + FinRobot strategy returned **17.87%**, compared with **17.09%** for FinRL-only and **5.19%** for SPY. The integrated strategy also improved Sharpe ratio from **1.93** to **2.07** and slightly reduced annualized volatility.

The most important lesson is that LLMs are more useful when they are converted into structured, auditable, and rule-based signals. FinRobot's value was not simply writing reports; it transformed research into portfolio multipliers. FinGPT's value was not replacing the trading engine; it helped identify when long exposure was more favorable.

Overall, our project shows that an AI trading system can be more effective when quantitative signals, research reasoning, and news sentiment are connected into one disciplined pipeline. The final strategy beat SPY, improved slightly over the FinRL-only baseline, and provided a more explainable investment process.